

Dong-A Univ. (ISPL)



동아대학교
DONG-A UNIVERSITY

Entropy & Decision Tree & KNN Method

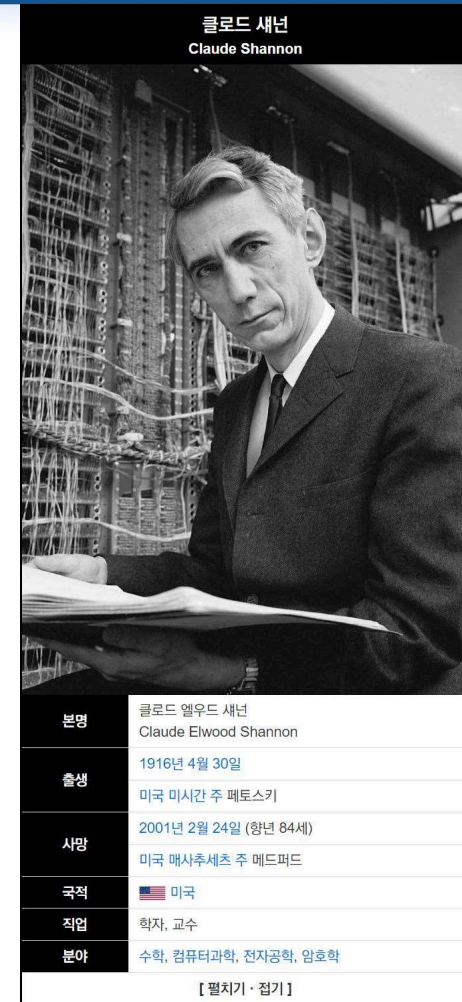
컴퓨터공학과
2025년 1학기 머신러닝

Entropy & Decision Tree

■ Entropy: measurement for uncertainty

- 불확실성(uncertainty) 정도를 나타내는 수치
- Shannon entropy: $H(X) = -\sum_{i=1}^n p_i \log_2 p_i$

Heads of coin	Tails of coin	Calculation	Entropy
50%	50%	$-(0.5 \times \log 0.5 + 0.5 \times \log 0.5) = 1$	1
100%	0%	$-(1.0 \times \log 1.0 + 0.0 \times \log 0.0) = 0$	0
90%	10%	$-(0.9 \times \log 0.9 + 0.1 \times \log 0.1) = 0.47$	0.47



Entropy & Decision Tree

- 불확실성이 높을수록 Entropy는 큰 값을 가짐
 - Binary Classification $0 \leq \text{Entropy} \leq 1$
 - 8-classes Classification $0 \leq \text{Entropy} \leq 3$
 - 16-classes Classification $0 \leq \text{Entropy} \leq 4$

Classification Type	Class number (n)	Maximum Entropy for Uniform Distribution H_{max}
Binary	2	$-2 \cdot \frac{1}{2} \log_2 \frac{1}{2} = 1$
8-classes	8	$-8 \cdot \frac{1}{8} \log_2 \frac{1}{8} = 3$
16-classes	16	$-16 \cdot \frac{1}{16} \log_2 \frac{1}{16} = 4$

Entropy & Decision Tree

- **Machine Learning에서 Entropy 활용 예**
 - [1] Deep Learning의 Loss Function
 - [2] Decision Tree
 - [3] Active Learning

Entropy & Decision Tree

Machine Learning에서 Entropy 활용 예 (MLP: Loss Function)

- [1] Deep Learning의 Loss Function : 학습 모델이 얼마나 잘못 예측하고 있는지는 표현하는 지표
 - 값이 낮을수록 모델이 정확하게 예측했다고 해석할 수 있음
 - Ex. Cross Entropy Error (CEE) 계산 방법

$$CEE(y, y') = - \sum_{i=1}^N y_i \times \log(y'_i)$$

- ❖ y: 정답 값
- ❖ y': 예측 값



$$h(x) = - \sum_{i=1}^n (p_i \log_2(p_i))$$

0	1	2	3	4	5	6	7	8	9
0	0	1	0	0	0	0	0	0	0

정답 값 (y, one-hot)

Model A의 예측 결과

0	1	2	3	4	5	6	7	8	9
0	0	0.8	0	0	0	0.1	0	0.1	0

예측 확률 (y') **CEE = 0.2231**

$$CEE(y, y') = -(1 \times \log(0.8)) = 0.2231$$

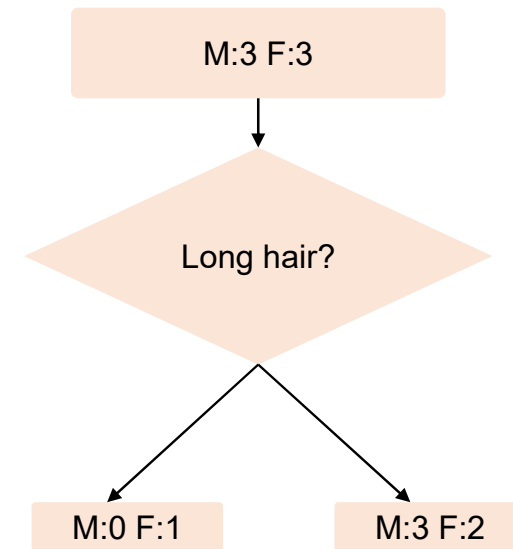
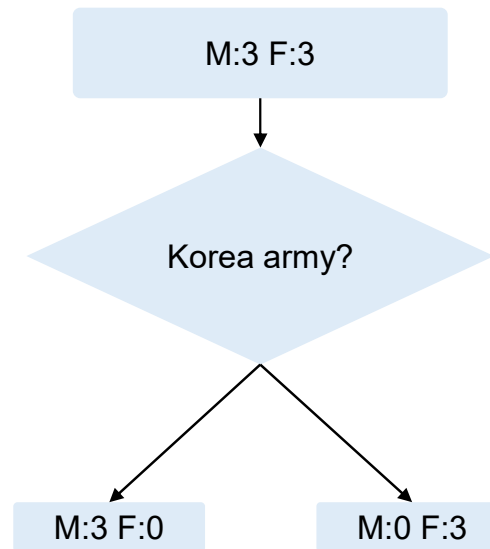
Entropy & Decision Tree

Machine Learning에서 Entropy 활용 예 (Decision Tree, DT)

[2] Decision Tree

- DT에서 **확실히 구분이 되는 특징**을 먼저 구분해 주는 것이 중요
- 확실히 구분이 되는 특징은 **불확실성(엔트로피)**가 작다는 것을 의미

Person	Korea army ?	Long hair?	Gender
1	Yes	No	Male
2	No	No	Female
3	Yes	No	Male
4	No	Yes	Female
5	Yes	No	Male
6	No	No	Female

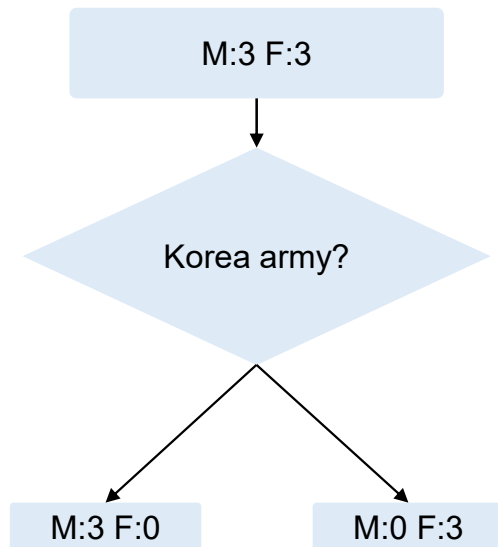


Entropy & Decision Tree

Machine Learning에서 Entropy 활용 예 (Decision Tree, DT)

[2] Decision Tree

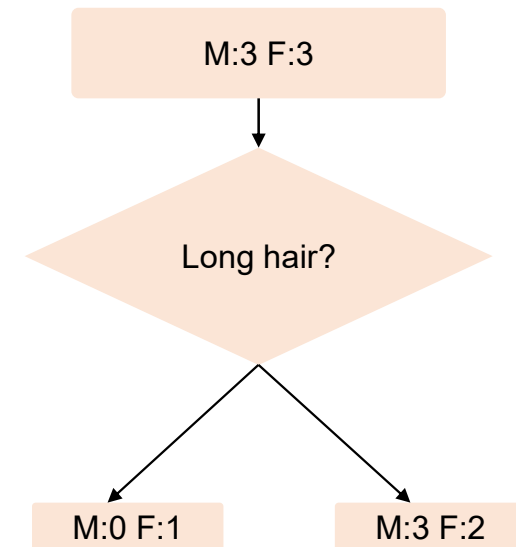
- DT에서 **확실히 구분이 되는 특징**을 먼저 구분해 주는 것이 중요
- 확실히 구분이 되는 특징은 **불확실성(엔트로피)**가 작다는 것을 의미



$$h(x) = - \sum_{i=1}^n (p_i \log_2(p_i))$$

$$\frac{3}{6} * ((-\frac{3}{3}) * \log(\frac{3}{3}) - (-\frac{0}{3}) * \log(\frac{0}{3})) + \frac{3}{6} * ((-\frac{3}{3}) * \log(\frac{3}{3}) - (-\frac{0}{3}) * \log(\frac{0}{3}))$$

totoal Entropy : **0**



$$\frac{1}{6} * ((-\frac{0}{1}) * \log(\frac{1}{0}) - (-\frac{1}{1}) * \log(\frac{1}{1})) + \frac{5}{6} * ((-\frac{3}{5}) * \log(\frac{3}{5}) - (-\frac{2}{5}) * \log(\frac{2}{5}))$$

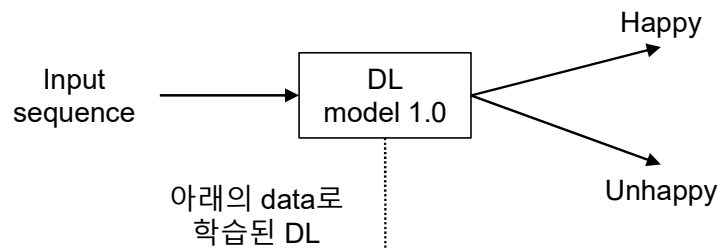
totoal Entropy : **0.966**

Entropy & Decision Tree

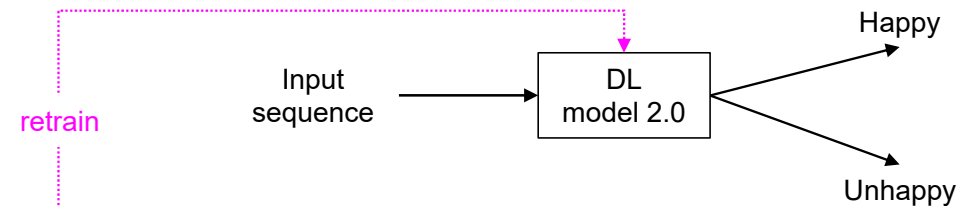
Machine Learning에서 Entropy 활용 예

- [3] Active Learning

➤ $h(x) = -\sum_{i=1}^n (p_i \log_2(p_i))$



	Happy	Unhappy	Entropy	
I love you	0.99	0.01	0.08	
I am angry	0.05	0.95	0.29	uncertain
I am sad	0.3	0.7	0.88	certain
I am feeling blue	0.6	0.4	0.97	



	Happy	Unhappy
I love you	0.99	0.01
I am angry	0.05	0.95
I am sad	0.01	0.99
I am feeling blue	0.04	0.96

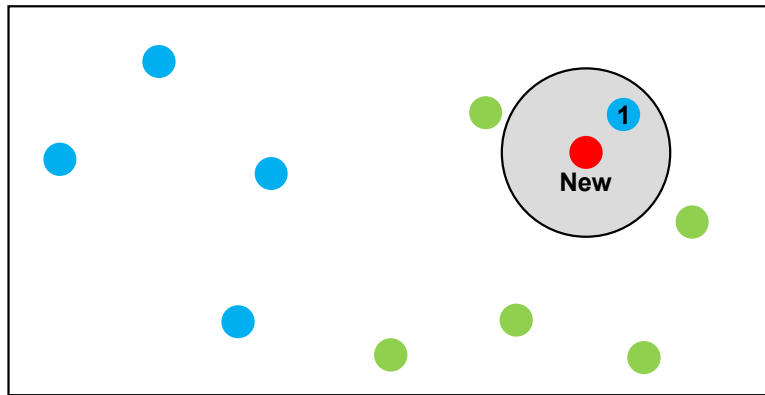
K-Nearest Neighbor (KNN)

▪ Supervised Learning: Model-based Learning

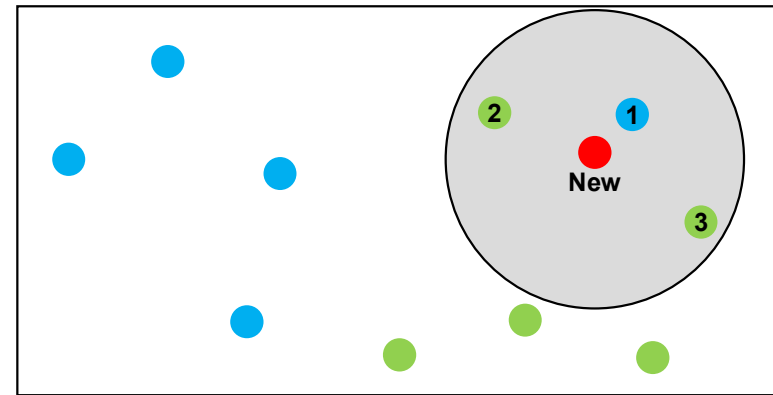
- Linear/Ridge/Lasso/Elastic Regression
- Deep Learning(MLP & CNN)
- Support Vector Machine
- Decision Tree
- KNN

▪ Unsupervised Learning

- K-means Algorithm): [Memory-based Learning] or [Lazy Learning]



K = 1 예시

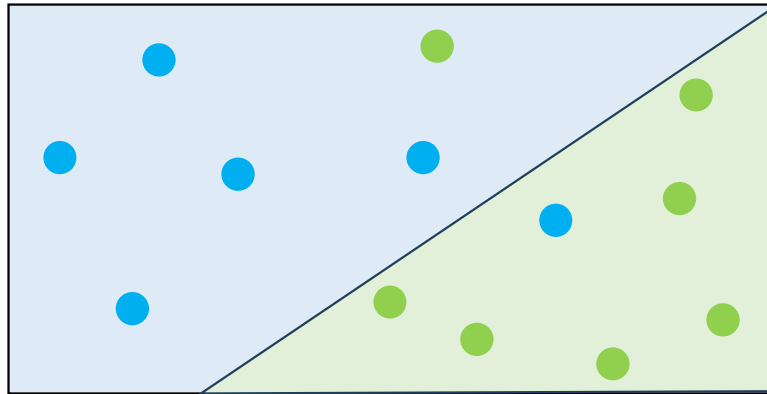


K = 3 예시

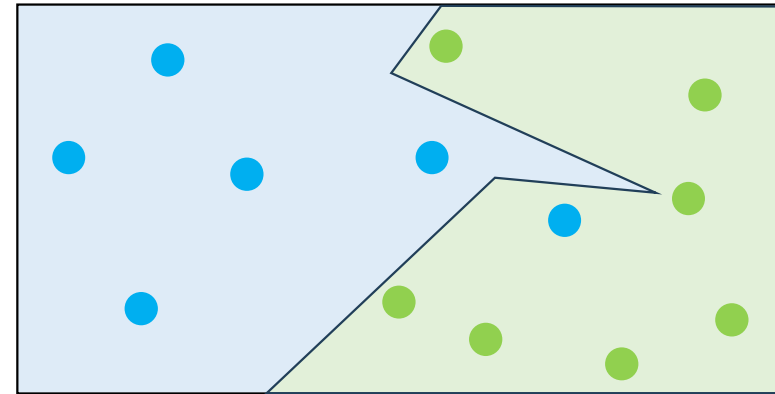
K-Nearest Neighbor (KNN)

▪ KNN Algorithm

- Linear vs non Linear



K = 1 (Linear boundary)



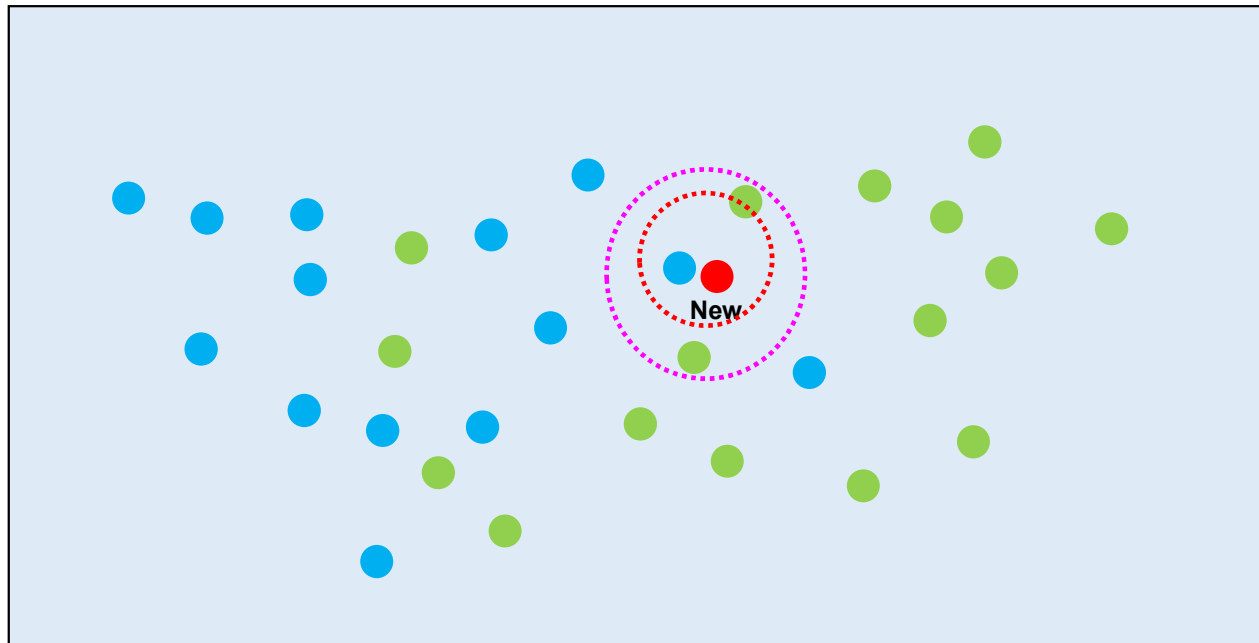
K = 3 (non Linear boundary)

- KNN 응용: (1) KNN 분류, (2) KNN 추정

K-Nearest Neighbor (KNN)

▪ KNN 분류

- 인접한 K개의 data로부터 Majority voting
 - K = nearest neighbors
 - K = 1경우 : 빨간 점선
 - K = 3경우 : 핑크 점선



K-Nearest Neighbor (KNN)

▪ KNN 분류

- 인접한 K개의 data로부터 Majority voting

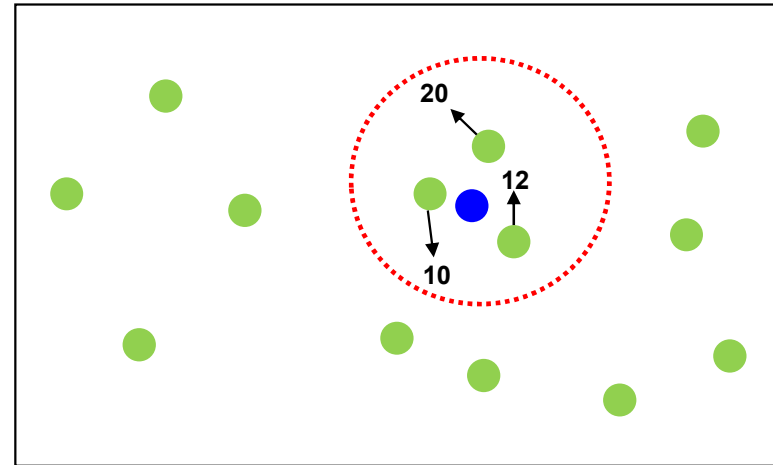
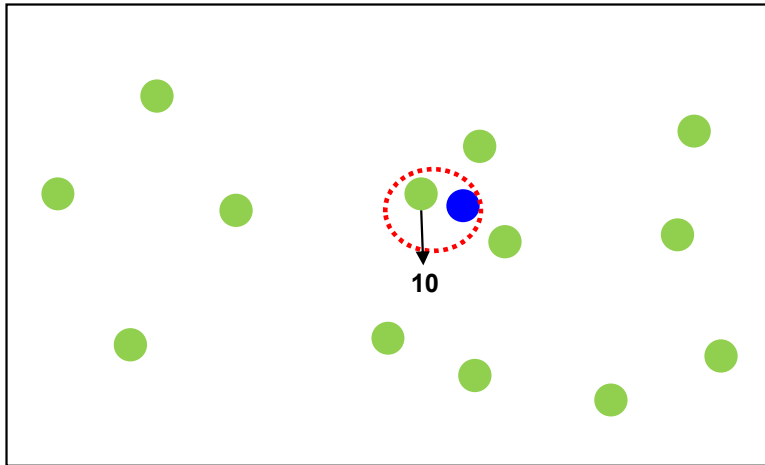
I	감기에 대한 정보				환자 상태	I
사람	기침	콧물	가래	발열	감기 유무	새로운 관측치와 거리
A	1.85	3.4	4.12	2.95	정상	1.54
B	2.9	3.2	3.77	3.1	정상	0.76
C	2.35	2.95	5.25	3.48	정상	2.00
D	3.7	3.8	4.05	3.85	감기	0.78
E	3.45	2.9	2.95	4.1	감기	1.28
F	3.95	2.6	3.4	4.2	감기	1.31
G	3.05	3.1	3.95	3.7	?	

K = 1 일 때, 정상
K = 3 일 때, 감기

K-Nearest Neighbor (KNN)

▪ KNN 분류

- 인접한 K개의 data로부터 평균/중간 값/Min/Max 중 택
 - K = number of nearest neighbors
 - K = 1 : new = 15
 - K = 3 : new = $(10+12+20)/3 = 14$



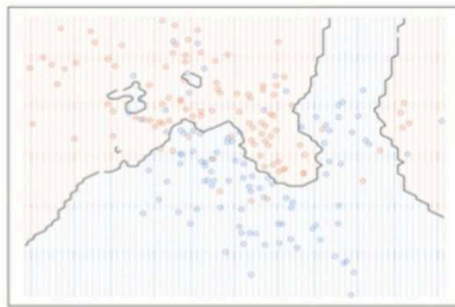
K-Nearest Neighbor (KNN)

▪ KNN Algorithm 이슈

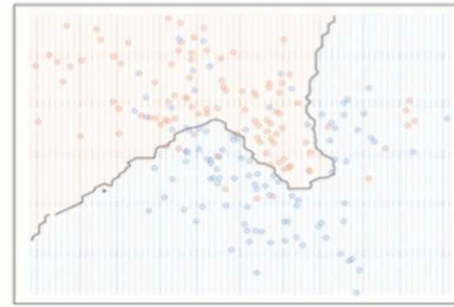
- [1] 최적의 K 를 어떻게 결정할 것인가? → 인접한 학습 data를 몇 개까지 탐색할 것인가?
($1 \leq K \leq$ 전체 data 개수 → Overfitting vs Underfitting)



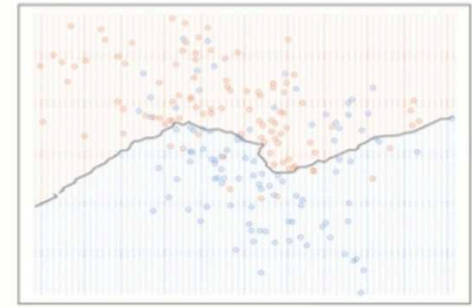
1-nearest neighbor



5-nearest neighbor



15-nearest neighbor



50-nearest neighbor

K-Nearest Neighbor (KNN)

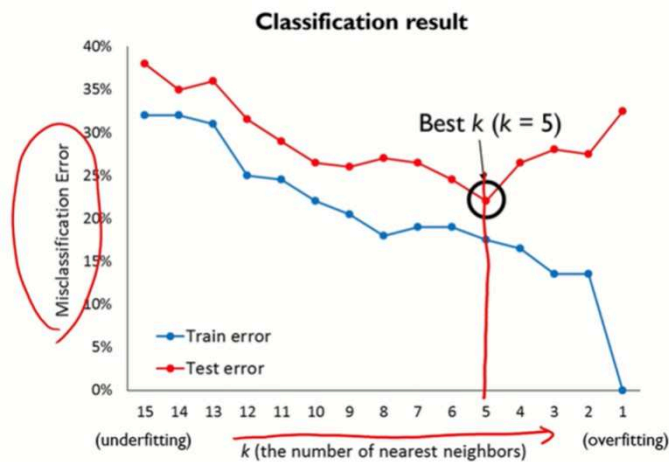
▪ KNN Algorithm 이슈

- [1] 최적의 K 를 어떻게 결정할 것인가? → 인접한 학습 data를 몇 개까지 탐색할 것인가?

- 분류모델: $MisclassError_k = \frac{1}{k} \sum_{i=1}^k I(c_i \neq \hat{c}_i)$ for $k = 1, 2, \dots, k^*$

$I(\cdot)$: Indicator Function

- $SSE_k = \sum_{i=1}^k (y_i - \hat{y}_i)^2$ for $k = 1, 2, \dots, k^*$



K-Nearest Neighbor (KNN)

▪ KNN Algorithm 이슈

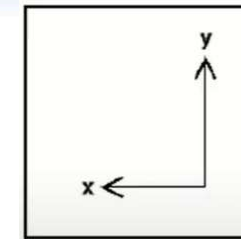
- [2] Data간 거리는 어떻게 측정할 것인가? → Distance Measurements

➤ L1 Norm (Manhattan Distance): $D_{Manhattan}(X,Y) = \sum_{i=1}^n |x_i - y_i|$

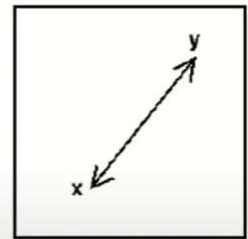
➤ L2 Norm (Euclidean Distance): $d_{(A,B)} = \sqrt{(a_1 - b_1)^2 + \dots + (a_p - b_p)^2} = \sqrt{\sum_{i=1}^p (a_i - b_i)^2}$

➤ Mahalanobis: $d_{Mahalanobis}(X,Y) = \sqrt{(X - Y)^T \Sigma^{-1} (X - Y)}$, Σ^{-1} : inverse of covariance matrix

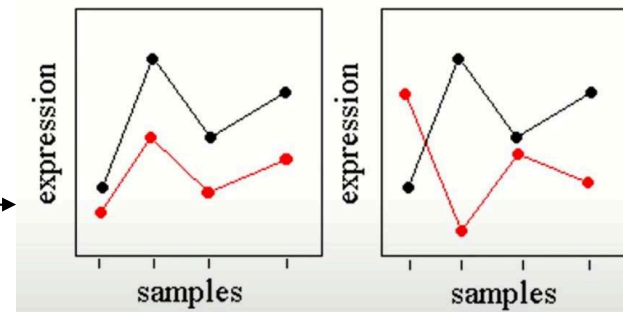
➤ Correlation Distance: $d_{corr}(X,Y) = 1 - r$, where $r = \sigma_{XY}$



Manhattan



Euclidean





Questions & Answers

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